

Submission DSA1-MID-SUB08

Question DSA1-MID-Q01

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visible work:

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.... ?

last line is hard to read; maybe path $4 \rightarrow 9 \rightarrow 6$; parent=6

Question DSA1-MID-Q02

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visible work:

Maintain a hash set of values seen so far.... ?

last line is hard to read; maybe hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant

Question DSA1-MID-Q03

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visible work:

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$?

last line is hard to read; maybe build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree

Question DSA1-MID-Q04

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visible work:

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.... ?

last line is hard to read; maybe A,B,C,D,E; BFS explores by distance layers and yields fewest edges

Question DSA1-MID-Q05

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visible work:

Invariant: the prefix before index i is sorted and contains the original prefix values.... ?

last line is hard to read; maybe sorted prefix invariant; initialization, insertion maintenance, termination imply correctness

Question DSA1-MID-Q06

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visible work:

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.... ?

last line is hard to read; maybe path $4 \rightarrow 9 \rightarrow 6$; parent=6

Question DSA1-MID-Q07

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visible work:

Maintain a hash set of values seen so far.... ?

last line is hard to read; maybe hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant

Question DSA1-MID-Q08

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visible work:

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$?

last line is hard to read; maybe build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree